

Medical Image Processing M2 Data Science & AI 2024-2025

www.sop.inria.fr/teams/asclepios/cours/MS_DS/

X. Pennec

Statistics on Riemannian manifolds and Lie groups



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Medical Image Analysis – M2 DSAI 2024-2025

Thursday afternoon

Course notes : http://www.sop.inria.fr/teams/asclepios/cours/MS_DS/

- Thur. Jan. 09, 14:00-17:00 [Xavier Pennec] Introduction to Medical Image Acquisition
- Thur. Jan. 16 13:30-16:30 [Xavier Pennec] Medical Image Registration
- Thur. Jan. 23 13:30-16:30 [Hervé Delingette] Image Filtering
- Thur. Jan. 30 13:30-16:30 [Xavier Pennec] Statistics on Riemannian manifolds and Lie groups
- Thur. Feb. 6 13:30-16:30 [Hervé Delingette] Principles and Metrics for Image segmentation
- Thur. Feb. 13 13:30-16:30 [Hervé Delingette] Image segmentation based on clustering
- Thur. Feb. 27 13:30-16:30 [Xavier Pennec] Diffeomorphic Registration and Computational Anatomy
- Thur. March 6, 13:30-16:30 [Hervé Delingette] Connexity and Shape Constrained Image segmentation
- Thur. March 13, 13:30-15:00, [XP & HD] EXAM

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Course overview

Introduction

- Why do we need statistics on manifolds?

Simple statistics on interesting manifolds

- The Riemannian Geometric framework
- Simple Statistics
 - Mean, Covariance, Gaussian, t-tests

Application to registration

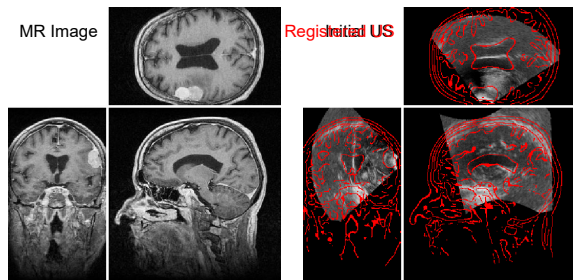
- Statistics on spine shapes
- Evaluation of registration accuracy

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Per-operative registration of MR/US images



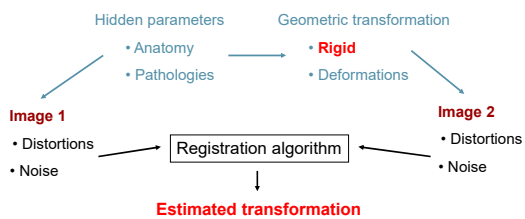
Performance Evaluation?

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Variability of a registration algorithm



Quantify the statistical Variability of the transformation:

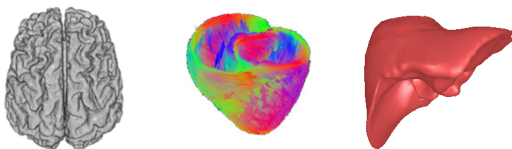
- Expected value (bias)
- Covariance matrix, std dev. (accuracy, precision)
 - On the transformation (rotation σ_r [rad], translation σ_t [mm])
 - Propagate on target points (TRE σ_x)

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Computational Anatomy



Computational Anatomy, an emerging discipline, P. Thompson, M. Miller, NeuroImage special issue 2004
Mathematical Foundations of Computational Anatomy, X. Pennec, S. Joshi, MICCAI workshop, 2006

Modeling and Analysis of the Human Anatomy

- Estimate representative / average organ anatomies
- Model organ development across time
- Establish normal variability
- To detect and classify of pathologies from structural deviations
- To adapt generic (atlas-based) to patients-specific models

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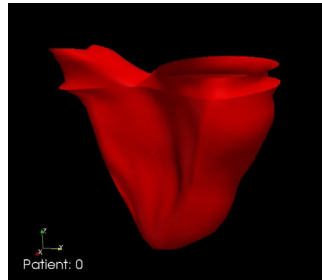
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Methods of computational anatomy

Remodeling of the right ventricle of the heart in tetralogy of Fallot

- Mean shape
- Shape variability
- Correlation with clinical variables
- Predicting remodeling effect

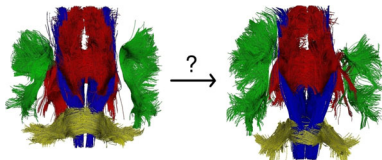


Shape of RV in 18 patients

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Shapes: forms & deformations



“Shape space” embedding [Kendall]

- Shape = what remains from the object when we remove all transformations from a given group
 - Transformation (rigid, similarity, affine) = nuisance factor
 - Shape manifold = quotient of the Object manifold by the group action
- Quotient spaces are non-linear (e.g. $\mathbb{R}^n$ / scaling = S^n)
- Kendall size & shape space: $(\mathbb{R}^n)^d / SO_n$

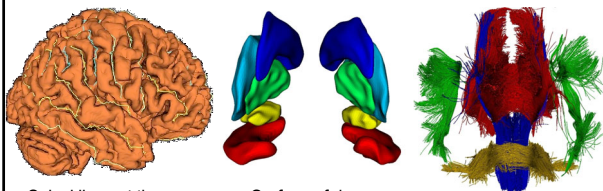
Statistics on shape spaces?

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Anatomical structures segmented in Brain Images



Sulcal lines at the surface of the cortex

Surface of deep brain structures

Fiber tracts from DTI

How to measure the variability across subjects?
Generic framework to deal with all object types?

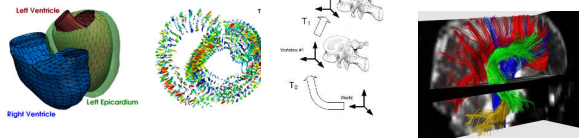
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Statistical Analysis of Geometric Features

Geometric features belong to manifolds

- Curves, tracts
- Surfaces
- Tensors, covariance matrices
- Transformations / deformations



Algorithms for statistics on geometric manifolds

- Definition of mean / covariance / PCA / distributions of geometric features?
- Mathematical structure = algorithmic bases

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Basic probabilities and statistics

Measure: random vector $\mathbf{x}$ of pdf $p_{\mathbf{x}}(z)$

Approximation: $\mathbf{x} \sim (\bar{\mathbf{x}}, \Sigma_{\mathbf{x}})$

- Mean: $\bar{\mathbf{x}} = E(\mathbf{x}) = \int \mathbf{z} \cdot p_{\mathbf{x}}(z) \cdot dz$
- Covariance: $\Sigma_{\mathbf{x}} = E[(\mathbf{x} - \bar{\mathbf{x}}) \cdot (\mathbf{x} - \bar{\mathbf{x}})^T]$

Propagation: $\mathbf{y} = h(\mathbf{x}) \sim \left(h(\bar{\mathbf{x}}), \frac{\partial h}{\partial \mathbf{x}} \cdot \Sigma_{\mathbf{x}} \cdot \frac{\partial h^T}{\partial \mathbf{x}} \right)$

Noise model: additive, Gaussian...

Principal component analysis

Statistical distance: Mahalanobis and χ^2

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Some problems with geometric features

Extrinsic means of 3D rotations: invariance w.r.t. the chart

$$\underline{\mathbf{R}} = \frac{1}{n} \sum_i \mathbf{R}_i \quad \underline{\mathbf{q}} = \frac{1}{n} \sum_i \mathbf{q}_i \quad \underline{\mathbf{r}} = \frac{1}{n} \sum_i \mathbf{r}_i$$

Noise on 3D rotations: invariance w.r.t. the transformation group

$$\Sigma_1 = \Sigma \quad \Sigma_2 = \Sigma \quad \Sigma_1' = \frac{8}{\pi^2} \Sigma \quad \Sigma_2' = \frac{\pi^2}{8} \Sigma$$

$$\Sigma = \sigma^2 \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

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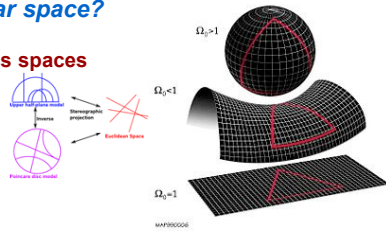
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Which non-linear space?

Constant curvatures spaces

- Sphere,
- Euclidean,
- Hyperbolic



Homogeneous spaces, Lie groups and symmetric spaces

Riemannian or affine connection spaces

Towards non-smooth quotient and stratified spaces

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Definition of a manifold

Intuitive idea

- "A manifold is a topological space which is locally Euclidean"
- "A Manifold is a topological space for which the neighborhood of each point is homeomorphic to the n dimensional Euclidean space"
 - Homeomorphism: F is bijective and F and F^{-1} are continuous (no folding or tearing transformation)

Definition: C^k manifold of dimension n ($n \geq 1$, $k \geq 1$, or $k = \infty$)

- A topological space M , together with a C^k atlas on M .
- Any equivalence class of atlases is called a *differentiable structure* of class C^k (and dimension n)

Differentiable manifolds

- When $k = \infty$, we say that M is a *smooth manifold*
- When $k=1$, M is a *differential manifold*
- When $k=0$, M is a *topological manifold*

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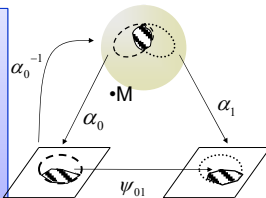
Traditional atlas definition

Given: Manifold M
Construct: Atlas A
Chart

- Region U_c in M (open disk)
- Region c in $\mathbb{R}^n$ (open disk)
- Function α_c taking U_c to c
 - Inverse

Atlas is collection of charts

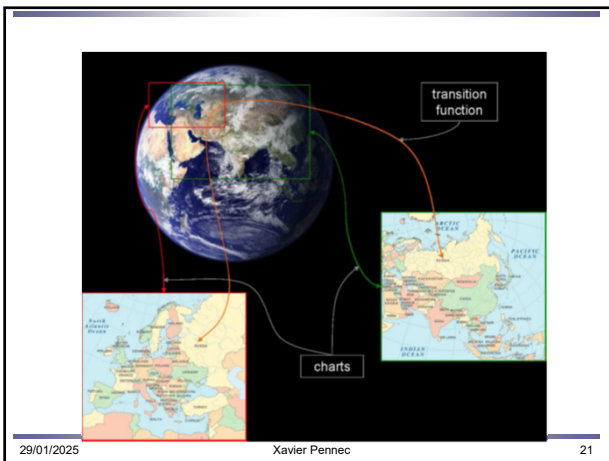
- Every point in M in at least one chart
- Overlap regions
- Transition functions:
 - $\psi_{01} = \alpha_1 \circ \alpha_0^{-1}$ smooth



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Differentiable manifolds

Computing on a manifold

- Extrinsic
 - Embedding in $\mathbb{R}^n$
- Intrinsic
 - Coordinates : charts
- Measuring?
 - Lengths
 - Straight lines
 - Volumes

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Measuring extrinsic distances

Basic tool: the scalar product

$$\langle v, w \rangle = v^t w$$

- Norm of a vector

$$\|v\| = \sqrt{\langle v, v \rangle}$$
- Length of a curve

$$L(\gamma) = \int \|\dot{\gamma}(t)\| dt$$

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Measuring extrinsic distances

Basic tool: the scalar product

$\langle v, w \rangle_p = v^t G(p) w$

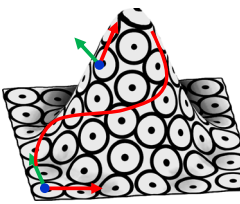
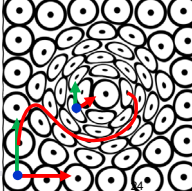
• Norm of a vector

$$\|v\|_p = \sqrt{\langle v, v \rangle_p}$$

• Length of a curve

$$L(\gamma) = \int \|\dot{\gamma}(t)\| dt$$


Bernhard Riemann
1826-1866

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Riemannian manifolds

Basic tool: the scalar product

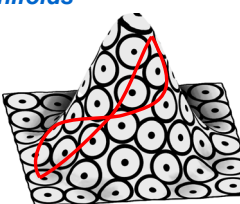
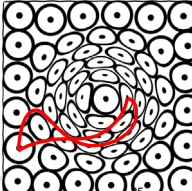
$\langle v, w \rangle_p = v^t G(p) w$

• Geodesics

- Shortest path between 2 points
- Calculus of variations (E.L.) :
2nd order differential equation (specific to the manifold)
- Free parameters: initial speed and starting point



Bernhard Riemann
1826-1866

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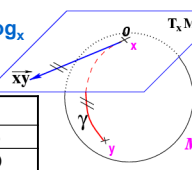
Bases of Algorithms in Riemannian Manifolds

Exponential map (Normal coordinate system):

- Exp_x = geodesic shooting parameterized by the initial tangent
- Log_x = unfolding the manifold in the tangent space along geodesics
 - Geodesics = straight lines with Euclidean distance
 - Geodesic completeness: covers $M \setminus \text{Cut}(x)$

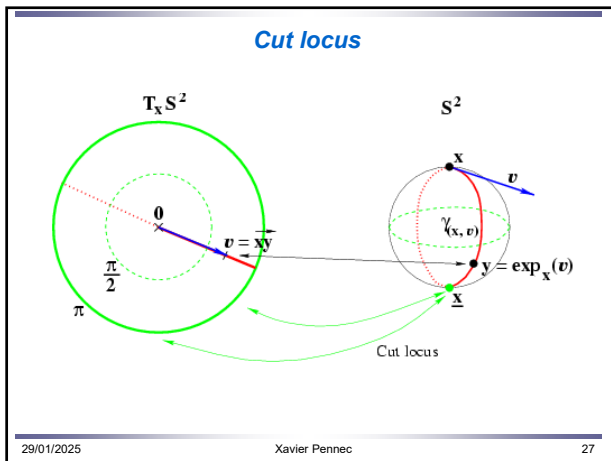
Reformulate algorithms with exp_x and log_x

Vector $\rightarrow$ Bi-point (no more equivalence classes)



Operation	Euclidean space	Riemannian
Subtraction	$xy = y - x$	$xy = \text{Log}_x(y)$
Addition	$y = x + xy$	$y = \text{Exp}_x(xy)$
Distance	$\text{dist}(x, y) = \ y - x\ $	$\text{dist}(x, y) = \ xy\ $
Gradient descent	$x_{i+1} = x_i - \epsilon \nabla \ell(x_i)$	$x_{i+1} = \text{Exp}_{x_i}(-\epsilon \nabla \ell(x_i))$

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Metric choice

Transformations (Lie group):

- Left (or right) invariant $\text{dist}(g, h) = \text{dist}(f \circ g, f \circ h) = \|f^{(-1)} \circ g\|_{Id}$
- Practical computations $\exp_f(\vec{\delta f}) = f \circ \vec{\delta f} \quad \vec{fg} = f^{(-1)} \circ g$
- No bi-invariant metric

Homogeneous manifolds $\text{dist}(x, y) = \text{dist}(g * x, g * y)$

- Invariance wrt the isotropy group
- Practical computations $\exp_x(\vec{\delta x}) = f_x * \vec{\delta x} \quad \vec{xy} = f_x^{(-1)} * \vec{y}$

General Riemannian manifolds

- Exp and log through numerical optimization / integration

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Example on 3D rotations

Space of rotations SO(3):

- Manifold: $R^t R = Id$ and $\det(R) = +1$
- Lie group:
 - Composition: $R_1 \circ R_2 = R_1 R_2$
 - Inversion: $R^{(-1)} = R^t$

Tangent space

- At Identity (skew symmetric matrices)
- At any point by left or right translation

Metrics on SO(3)

- Left / right invariant metrics
- Induced by the ambient space: bi-invariance

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Example on 3D rotations

Group exponential

- One parameter subgroups
- Matrix exponential and Rodrigue's formula

Exponential map for the bi-invariant metric

- Geodesic starting at identity = one parameter subgroups
- Geodesic everywhere by left (or right) translation

More details in the memo on rotations on the web

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Noise model: additive, Gaussian...

Principal component analysis

Statistical distance: Mahalanobis and χ^2

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Random variable in a Riemannian Manifold

Intrinsic pdf of $\mathbf{x}$

- For every set H

$$P(\mathbf{x} \in H) = \int_H p(y) dM(y)$$



- Lebesgue's measure

→ Uniform Riemannian Measure $dM(y) = \sqrt{\det(G(y))} dy$

Expectation of an observable in M

- $E_x[\phi] = \int_M \phi(y) p(y) dM(y)$
- $\phi = \text{dist}^2$ (variance) : $E_x[\text{dist}(\cdot, y)^2] = \int_M \text{dist}(y, z)^2 p(z) dM(z)$
- $\phi = \log(p)$ (information) : $E_x[\log(p)] = \int_M p(y) \log(p(y)) dM(y)$
- $\phi = x$ (mean) : $E_*[\mathbf{x}] = \int_M y p(y) dM(y)$

Integral only valid in Hilbert/Wiener spaces [Fréchet 44]

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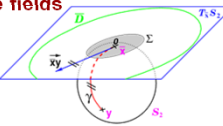
Statistical tools: Moments

Tensor moments of a random point on M

- $\mathfrak{M}_1(x) = \int_M \bar{x} \bar{z} dP(z)$ Tangent mean: (0,1) tensor field
- $\mathfrak{M}_2(x) = \int_M \bar{x} \bar{z} \otimes \bar{x} \bar{z} dP(z)$ (0,2) tensor field
- $\mathfrak{M}_k(x) = \int_M \bar{x} \bar{z} \otimes \bar{x} \bar{z} \otimes \dots \otimes \bar{x} \bar{z} dP(z)$ k-contravariant tensor field
- $\sigma^2(x) = \text{Tr}_g(\mathfrak{M}_2(x)) = \int_M \text{dist}^2(x, z) dP(z)$ Mean square distance

Tangent mean and tangent covariance fields

- Tg mean: $\mathfrak{M}_1(x) = \int_M \bar{x} \bar{z} dP(z)$
- Tg cov: $\text{Cov}(x) = \mathfrak{M}_2(x) - \mathfrak{M}_1(x) \otimes \mathfrak{M}_1(x)$



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Several definitions of the mean

Tensor moments of a random point on M

- $\mathfrak{M}_1(x) = \int_M \bar{x} \bar{z} dP(z)$ Tangent mean: (0,1) tensor field
- $\mathfrak{M}_2(x) = \int_M \bar{x} \bar{z} \otimes \bar{x} \bar{z} dP(z)$ (0,2) tensor field
- $\mathfrak{M}_k(x) = \int_M \bar{x} \bar{z} \otimes \bar{x} \bar{z} \otimes \dots \otimes \bar{x} \bar{z} dP(z)$ k-contravariant tensor field
- $\sigma^2(x) = \text{Tr}_g(\mathfrak{M}_2(x)) = \int_M \text{dist}^2(x, z) dP(z)$ Mean square distance

Fréchet mean set

- Fréchet mean [1948] = global minima of MSD
 - Karcher mean [1977] = local minima of MSD
 - Exponential barycenters [Emery & Mokobodzki 1991]
- $\mathfrak{M}_1(\bar{x}) = \int_M \bar{x} \bar{z} dP(z) = 0$ (critical points if $P(C) \neq 0$)
sometimes called Riemannian center of mass
[Groove & Karcher 1973-1976] under uniqueness assumptions



Maurice Fréchet
(1878-1973)

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Fréchet expectation (1948)

Minimizing the variance

$$\mathbb{E}[\mathbf{x}] = \underset{y \in M}{\operatorname{argmin}} \left(\mathbb{E}[\operatorname{dist}(y, \mathbf{x})^2] \right)$$

Existence

- Finite variance at one point

Characterization as an exponential barycenter ($P(C)=0$)

$$\operatorname{grad}(\sigma_x^2(y)) = 0 \quad \Rightarrow \quad \mathbb{E}[\overline{\mathbf{x}\mathbf{x}}] = \int_M \overline{\mathbf{x}\mathbf{x}} p_x(z) dM(z) = 0$$

Uniqueness Karcher 77 / Kendall 90 / Afsari 10 / Le 10

- Unique Karcher mean (thus Fréchet) if distribution has support in a regular geodesic ball with radius $r < r^* = \frac{1}{2} \min(\operatorname{inj}(M), \pi/\sqrt{\kappa})$ (κ upper bound on sectional curvatures on M)
- Empirical mean: a.s. uniqueness [Arnaudon & Miclo 2013]

Other central primitives

$$\mathbb{E}^\alpha[\mathbf{x}] = \underset{y \in M}{\operatorname{argmin}} \left(\mathbb{E}[\operatorname{dist}(y, \mathbf{x})^\alpha] \right)^{1/\alpha}$$

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A gradient descent (Gauss-Newton) algorithm

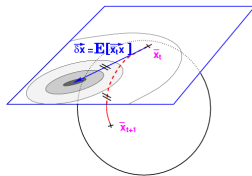
Vector space

$$f(x+v) = f(x) + \nabla f^T \cdot v + \frac{1}{2} v^T \cdot H_f \cdot v$$

$$x_{t+1} = x_t + v \quad \text{with} \quad v = -H_f^{(-1)} \cdot \nabla f$$

Manifold

$$f(\exp_x(v)) = f(x) + \nabla f(v) + \frac{1}{2} H_f(v, v)$$



$$\nabla(\sigma_x^2(y)) = -2 \mathbb{E}[\overline{\mathbf{y}\mathbf{x}}] = \frac{-2}{n} \sum_i \overline{\mathbf{y}\mathbf{x}_i}$$

$$H_{\sigma_t^2} \approx 2Id$$

Geodesic marching

$$\bar{x}_{t+1} = \exp_{\bar{x}_t}(v) \quad \text{with} \quad v = \mathbb{E}[\overline{\mathbf{y}\mathbf{x}}]$$

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Algorithms to compute the mean

Karcher flow (gradient descent)

$$\bar{x}_{t+1} = \exp_{\bar{x}_t}(\epsilon_t v_t) \quad \text{with} \quad v_t = \mathbb{E}(\overline{\mathbf{y}\mathbf{x}}) = \frac{1}{n} \sum_i \log_{\bar{x}_t}(x_i)$$

- Usual algorithm with $\epsilon_t = 1$ can diverge on SPD matrices [Bini & Iannazzo, Linear Algebra Appl., 438:4, 2013]
- Convergence for non-negative curvature (p-means) [Afsari, Tron and Vidal, SICON 2013]

Inductive / incremental weighted means

- $\bar{x}_{k+1} = \exp_{\bar{x}_k}(\frac{1}{k} v_k)$ with $v_k = \log_{\bar{x}_k}(x_{k+1})$
- On negatively curved spaces [Sturm 2003], BHV centroid [Billera, Holmes, Vogtmann, 2001]
- On non-positive spaces [G. Cheng, J. Ho, H. Salehian, B. C. Vemuri 2016]

Stochastic algorithm

- [Arnaudon & Miclo, Stoch. Processes and App. 124, 2014]

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Example on 3D rotations

Space of rotations $SO(3)$:

- Manifold: $R^T \cdot R = Id$ and $\det(R) = +1$
- Lie group ($R_1 \circ R_2 = R_1 R_2$ & Inversion: $R^{(-1)} = R^T$)

Metrics on $SO(3)$: compact space, there exists a bi-invariant metric

- Left / right invariant / induced by ambient space $\langle X, Y \rangle = \text{Tr}(X^T Y)$

Group exponential

- One parameter subgroups = bi-invariant Geodesic starting at Id
 - Matrix exponential and Rodrigue's formula: $R = \exp(X)$ and $X = \log(R)$
- Geodesic everywhere by left (or right) translation

$$\text{Log}_R(U) = R \log(R^T U) \quad \text{Exp}_R(X) = R \exp(R^T X)$$

Bi-invariant Riemannian distance

- $d(R, U) = \|\log(R^T U)\| = \theta(R^T U)$

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Example with 3D rotations

Principal chart: rotation vector: $r = \theta \cdot n$

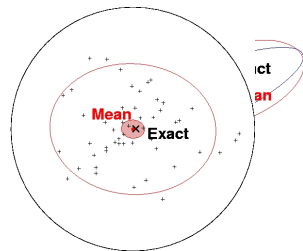
Distance: $\text{dist}(R_1, R_2) = \|r_1^{(-1)} \circ r_2\|$

Frechet mean:

$$\bar{R} = \arg \min_{R \in SO_3} \left(\sum_i \text{dist}(R, R_i) \right)$$

Centered chart:

mean = barycenter



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tPCA vs PGA

tPCA

- Generative model: Gaussian
- Find the subspace that best explains the variance
 - Maximize the squared distance to the mean

PGA (Fletcher 2004, Sommer 2014)

- Generative model:
 - Implicit uniform distribution within the subspace
 - Gaussian distribution in the vertical space
- Find a low dimensional subspace (geodesic subspaces?) that minimizes the error
 - Minimize the squared Riemannian distance from the measurements to that sub-manifold (no closed form)

Different models in curved spaces (no Pythagore thm)

Extension to BSA (Pennec 2018)

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Distributions for parametric tests

Uniform density:

- maximal entropy knowing X $p_x(z) = \text{Ind}_X(z) / \text{Vol}(X)$

Generalization of the Gaussian density:

- Stochastic heat kernel $p(x, y, t)$ [complex time dependency]
- Wrapped Gaussian [Infinite series difficult to compute]
- Maximal entropy knowing the mean and the covariance

$$N(y) = k \cdot \exp\left(\frac{\bar{x}^T \cdot \Gamma \cdot \bar{x}}{2}\right) \quad \Gamma = \Sigma^{(-1)} - \frac{1}{3} \text{Ric} + O(\sigma) + \varepsilon(\sigma/r)$$

$$k = (2\pi)^{-n/2} \cdot \det(\Sigma)^{-1/2} \cdot (1 + O(\sigma^3) + \varepsilon(\sigma/r))$$

Mahalanobis D2 distance / test:

$$\mu_x^2(y) = \bar{x}^T \cdot \Sigma_{xx}^{(-1)} \cdot \bar{x}$$

- Any distribution:

$$E[\mu_x^2(x)] = n$$

- Gaussian:

$$\mu_x^2(x) \propto \chi_n^2 + O(\sigma^3) + \varepsilon(\sigma/r)$$

[Pennec, JMV06, NSIP'99]

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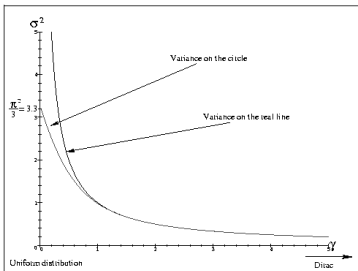
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Gaussian on the circle

Exponential chart: $x = r\theta \in]-\pi, r; \pi, r[$

Gaussian: truncated standard Gaussian



$r \rightarrow \infty$: standard Gaussian
(Ricci curvature $\rightarrow 0$)

$\gamma \rightarrow 0$: uniform pdf with
 $\sigma^2 = (\pi, r)^2 / 3$
(compact manifolds)

$\gamma \rightarrow \infty$: Dirac

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Computing on manifolds: a summary

The Riemannian metric easily gives

- Intrinsic measure and probability density functions
- Expectation of a function from M into R (variance, entropy)

Integral or sum in M : minimize an intrinsic functional

- Fréchet / Karcher mean: minimize the variance
- Filtering, convolution: weighted means
- Gaussian distribution: maximize the conditional entropy

The exponential chart corrects for the curvature at the reference point

- Gradient descent: geodesic walking
- Covariance and higher order moments
- Laplace Beltrami for free

[Pennec, NSIP'99, JMV 2006, Pennec et al, IJCV 66(1) 2006, Arsigny, PhD 2006]

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Course overview

Introduction

- Why do we need statistics on manifolds?

Simple statistics on interesting manifolds

- The Riemannian Geometric framework
- Simple Statistics
 - Mean, Covariance, Gaussian, t-tests

Application to registration

- Statistics on spine shapes
- Evaluation of registration accuracy

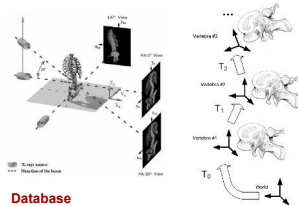
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Statistical Analysis of the Scoliotic Spine

[J. Boisvert, X. Pennec, N. Ayache, H. Labelle, F. Chéret, ISBI'06]



Database

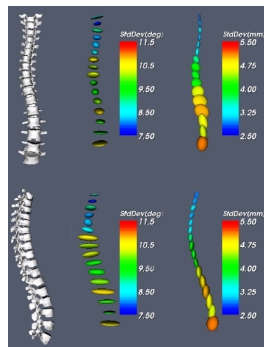
- 307 Scoliotic patients from the Montreal's Sainte-Justine Hospital.
- 3D Geometry from multi-planar X-rays

Mean

- Main translation variability is axial (growth?)
- Main rotation var. around anterior-posterior axis

PCA of the Covariance

- 4 first variation modes have clinical meaning

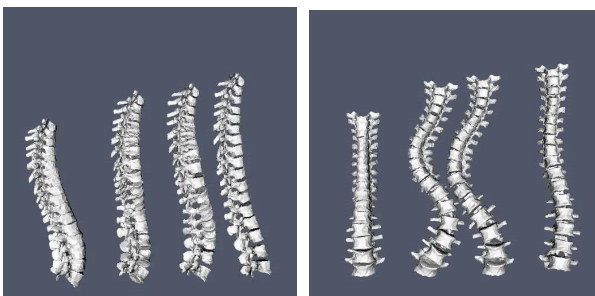


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Statistical Analysis of the Scoliotic Spine



- Mode 1: King's class I or III
- Mode 2: King's class I, II, III
- Mode 3: King's class IV + V
- Mode 4: King's class V (+II)

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MR-US Images

Pre - Operative MR Image

axial

coronal

sagittal

Per - Operative US Image

axial

coronal

sagittal

Acquisition of images : L. & D. Auer, M. Rudolf

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Typical Registration Result
with Bivariate Correlation Ratio

Pre - Operative MR Image

axial

coronal

sagittal

Per - Operative US Image

Registered

axial

coronal

sagittal

Acquisition of images : L. & D. Auer, M. Rudolf

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US Intensity
MR Intensity and Gradient

Multi-view

0 130 1302

0 134 1402

Y

93

X

83

Z

88

0

145

0.00

Multi-view

0 130 1302

0 134 1402

Y

93

X

83

Z

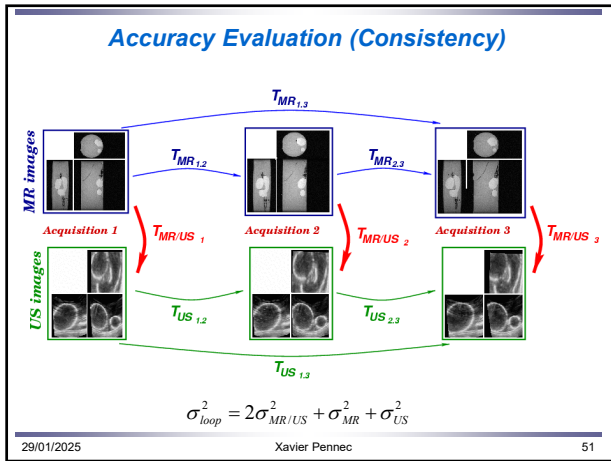
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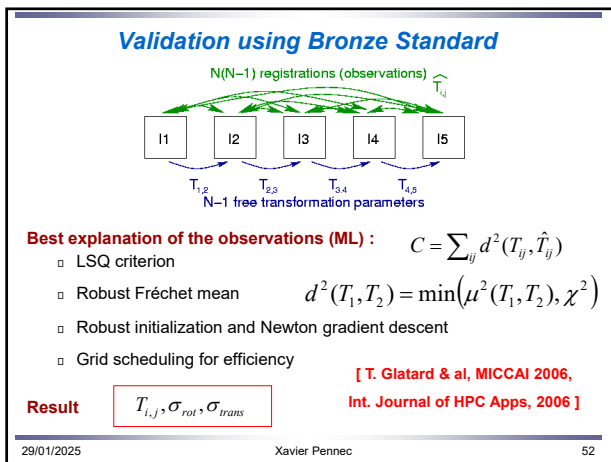
0

145

0.00

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Results on per-operative patient images

Data (per-operative US)

- 2 pre-op MR (0.9 x 0.9 x 1.1 mm)
- 3 per-op US (0.63 and 0.95 mm)
- 3 loops

Robustness and precision

	Success	var rot (deg)	var trans (mm)
MI	29%	0.53	0.25
CR	90%	0.45	0.17
BCR	85%	0.39	0.11

Consistency of BCR

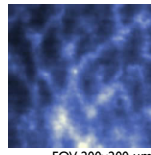
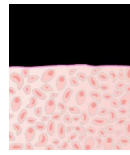
	var rot (deg)	var trans (mm)	var test (mm)
Multiple MR	0.06	0.06	0.10
Loop	2.22	0.82	2.33
MR/US	1.57	0.58	1.65

[Roche et al, TMI 20(10), 2001]
[Pennec et al, Multi-Sensor Image Fusion, Chap. 4, CRC Press, 2005]

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Mosaicing of Confocal Microscopic in Vivo Video Sequences.

Cellvizio: Fibered confocal fluorescence imaging



FOV 200x200 μm
FOV 2747x638 μm

Courtesy of Mike Booth, MGH, Boston, MA



[T. Vercauteren et al., MICCAI 2005, T.1, p.753-760]

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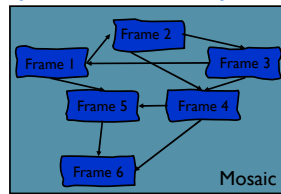
Mosaicing of Confocal Microscopic in Vivo Video Sequences.

Common coordinate system

- ▢ Multiple rigid registration
- ▢ Refine with non rigid

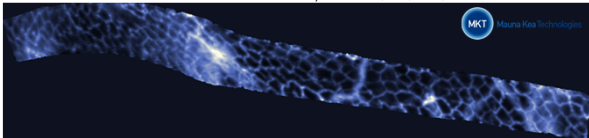
Mosaic image creation

- ▢ Interpolation / approximation with irregular sampling



Courtesy of Mike Booth, MGH, Boston, MA

FOV 2747x638 μm



[T. Vercauteren et al., MICCAI 2005, T.1, p.753-760]

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